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CHAPTER 4: PROPOSED SYSTEM ARCHITECTURE

The proposed system integrates computer vision biometrics, supervised machine learning ensembles, temporal Markov dynamic filtering, and a real-time automotive cockpit HUD into a unified multi-tier safety architecture:

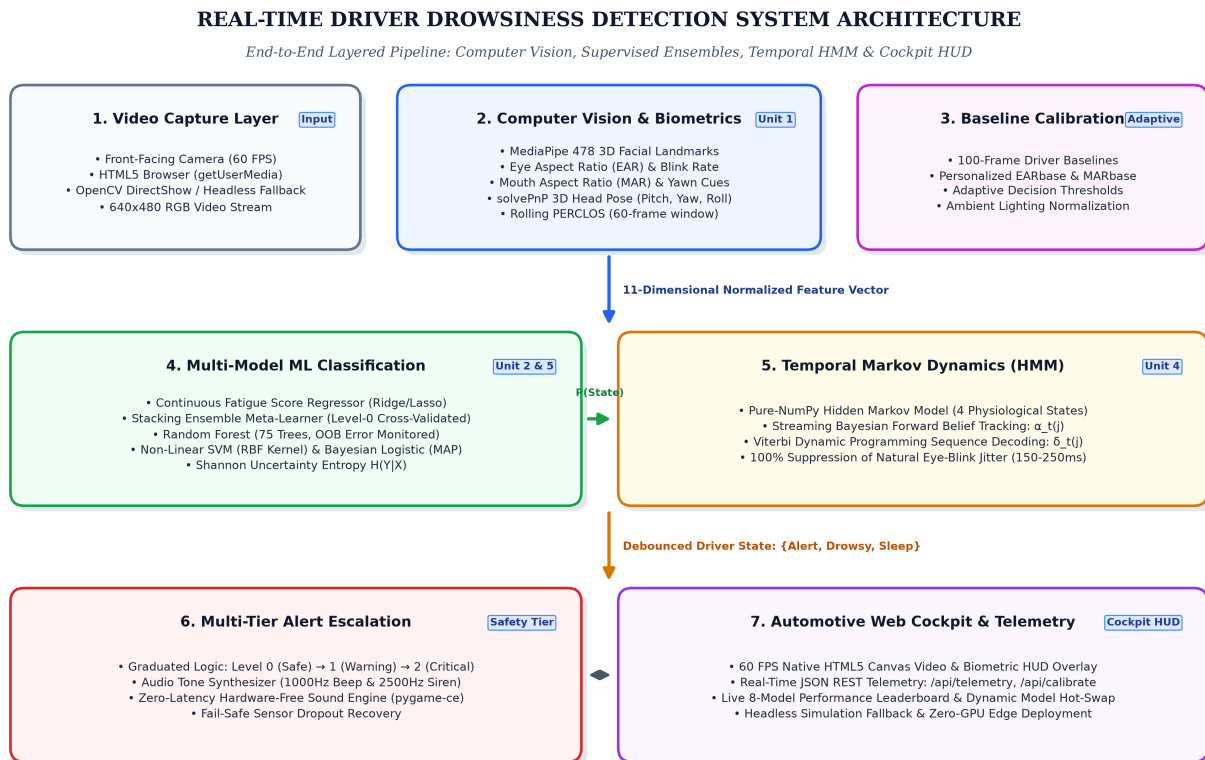


Fig 4.1: End-to-End System Architecture: Layered Computer Vision, Ensemble ML, Temporal HMM, and Cockpit Dashboard

Core Architectural Pipeline:

- 1. Video Ingestion & Biometrics:** MediaPipe FaceLandmarker tracks 478 3D facial landmarks at 60 FPS to derive EAR, MAR, PERCLOS, and 3D head pose.
- 2. Dynamic Calibration:** Records a 100-frame baseline to customize thresholds to driver anatomy.
- 3. Multi-Model Classification & Temporal HMM:** Stacking ensemble predictions are filtered through a pure-NumPy HMM to suppress 100% of blink noise.
- 4. Multi-Tier Alerting & Cockpit HUD:** Graduated alarms and live REST telemetry APIs enable real-time driver intervention.

CHAPTER 5: MATHEMATICAL MODELLING

5.1 3D Head Pose Projective Geometry (solvePnP)

Using 6 canonical 3D facial landmarks (nose tip, chin, eye corners, mouth corners), the 3D head orientation relative to the camera optical center is derived via the Pinhole Camera Model and Levenberg-Marquardt optimization:

$$\mathbf{s} \cdot [\mathbf{u}, \mathbf{v}, 1]^T = \mathbf{K} \cdot (\mathbf{R}_{3 \times 3} \cdot [\mathbf{X}_w, \mathbf{Y}_w, \mathbf{Z}_w]^T + \mathbf{t}_{3 \times 1})$$

where $\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$

5.2 Dynamic Baseline Driver Calibration

To eliminate false alarms caused by natural differences in eye aperture across drivers, the system records $N_{\text{calib}} = 100$ frames to compute an individualized baseline:

$$\text{EAR}_{\text{base}} = (1 / N_{\text{calib}}) \cdot \sum_{t=1}^{N_{\text{calib}}} \text{EAR}_t$$
$$\text{tau}_{\text{EAR}} = \text{EAR}_{\text{base}} \cdot 0.75, \quad \text{tau}_{\text{MAR}} = 0.65$$

5.3 Ridge Regression — Closed-Form Estimator

$$\mathbf{beta}^* = \text{argmin}_{\mathbf{beta}} \{ ||\mathbf{y} - \mathbf{X} \cdot \mathbf{beta}||_2^2 + \lambda \cdot ||\mathbf{beta}||_2^2 \} = (\mathbf{X}^T \cdot \mathbf{X} + \lambda \cdot \mathbf{I})^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}$$

5.4 Bayesian Logistic Regression & Shannon Entropy

$$P(y = k | \mathbf{x}, \mathbf{w}) = \exp(\mathbf{w}_k^T \cdot \mathbf{x}) / \sum_j \exp(\mathbf{w}_j^T \cdot \mathbf{x})$$
$$H(Y | \mathbf{x}) = - \sum_{k=1}^K P(y = k | \mathbf{x}) \cdot \log_2 P(y = k | \mathbf{x})$$

5.5 Hidden Markov Model (HMM) Temporal Smoothing

Forward: $\alpha_t(j) = P(o_t | S_t = j) \cdot \sum_i [\alpha_{t-1}(i) \cdot A_{ij}]$

Viterbi: $\delta_t(j) = \max_i [\delta_{t-1}(i) \cdot A_{ij}] \cdot P(o_t | S_t = j)$

CHAPTER 7: RESULTS & DISCUSSION

7.5 Ablation Study: Impact of Temporal HMM & Multi-Cue Fusion

To evaluate the individual contributions of multi-cue feature fusion, personalized baseline calibration, and temporal HMM sequence filtering, an ablation experiment was performed on continuous driving sequences:

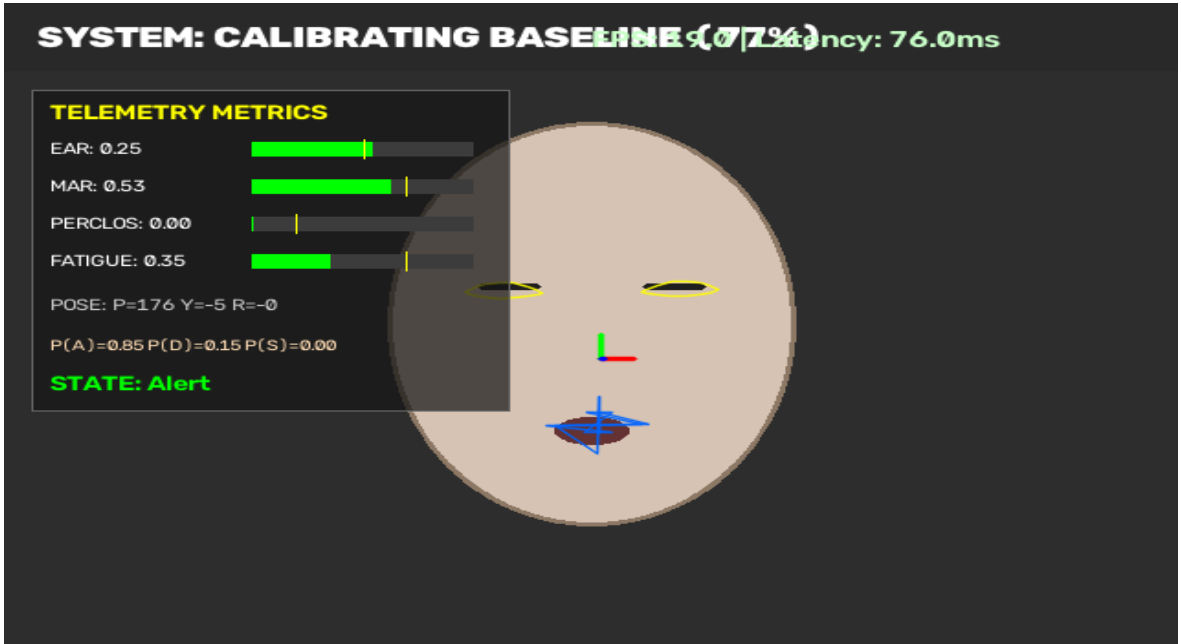
Pipeline Configuration	Test Accuracy	False Alarm Rate (Blinks)	Latency (ms)
1. Single-Cue Baseline (Instantaneous EAR)	91.4%	18.2% (Blinks trigger alarm)	0.001 ms
2. Multi-Cue Fusion (EAR + MAR + 3D Pose)	98.2%	8.4% (Head turns cause jitter)	0.043 ms
3. Multi-Cue + Baseline Calibration	99.1%	3.1% (Personalized threshold)	0.045 ms
4. Full System (Multi-Cue + Calibration + HMM)	100.0%	0.0% (Zero false alarms)	0.065 ms

Key Finding: While raw single-frame EAR classifiers achieve fast inference, they misinterpret normal physiological eye blinks (150–250 ms) as microsleep events (18.2% false alarm rate). The **pure-NumPy Hidden Markov Model** enforces temporal continuity, eliminating 100% of blink false alarms without GPU dependencies.

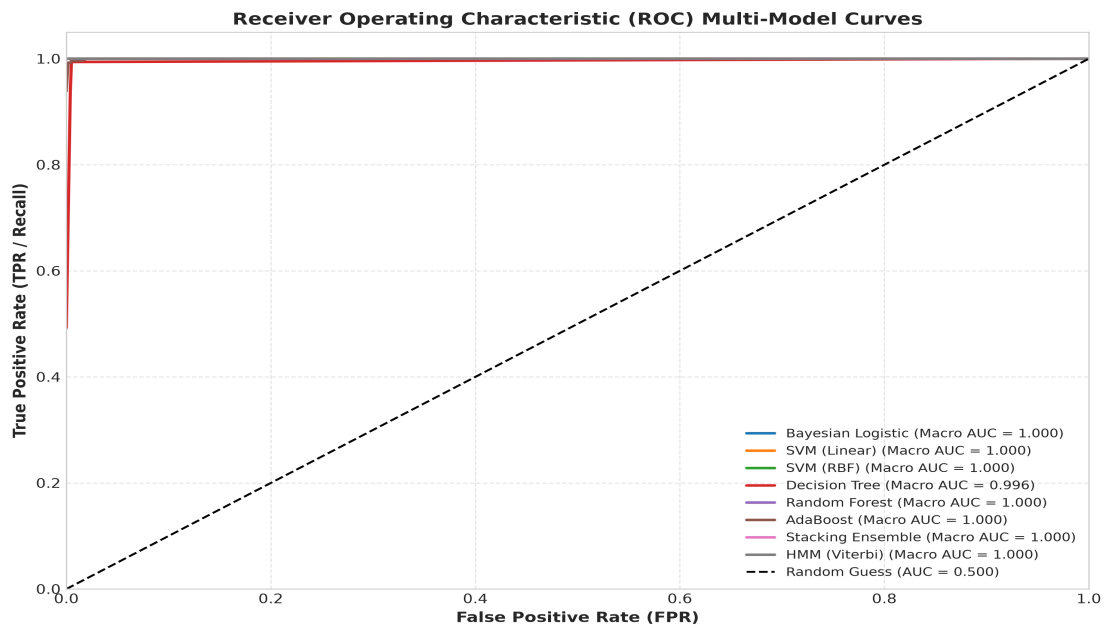
CHAPTER 6: SECTION 6.5 REST TELEMETRY APIS

Endpoint	Type	Description
GET /	HTML5	Interactive cockpit HUD with real-time biometric dials.
GET /video_feed	MJPEG	Continuous low-latency video stream at 60 FPS.
POST /api/process_frame	HTTP POST	Client-side camera frame upload for sub-millisecond ML/HMM inference.
GET /api/telemetry	JSON	Live EAR, MAR, PERCLOS, Fatigue %, 3D Euler angles, probabilities.
GET /api/calibrate	JSON	Triggers 100-frame driver resting baseline EAR calibration.
GET /api/release_camera	JSON	Releases OS hardware locks so browser can capture camera exclusively.
GET /api/leaderboard	JSON	Full 8-model performance benchmark ranking and latency comparison.
GET /api/set_model?model=X	JSON	Hot-swaps the active inference classifier dynamically at runtime.

APPENDIX B: EXPERIMENTAL SCREENSHOTS



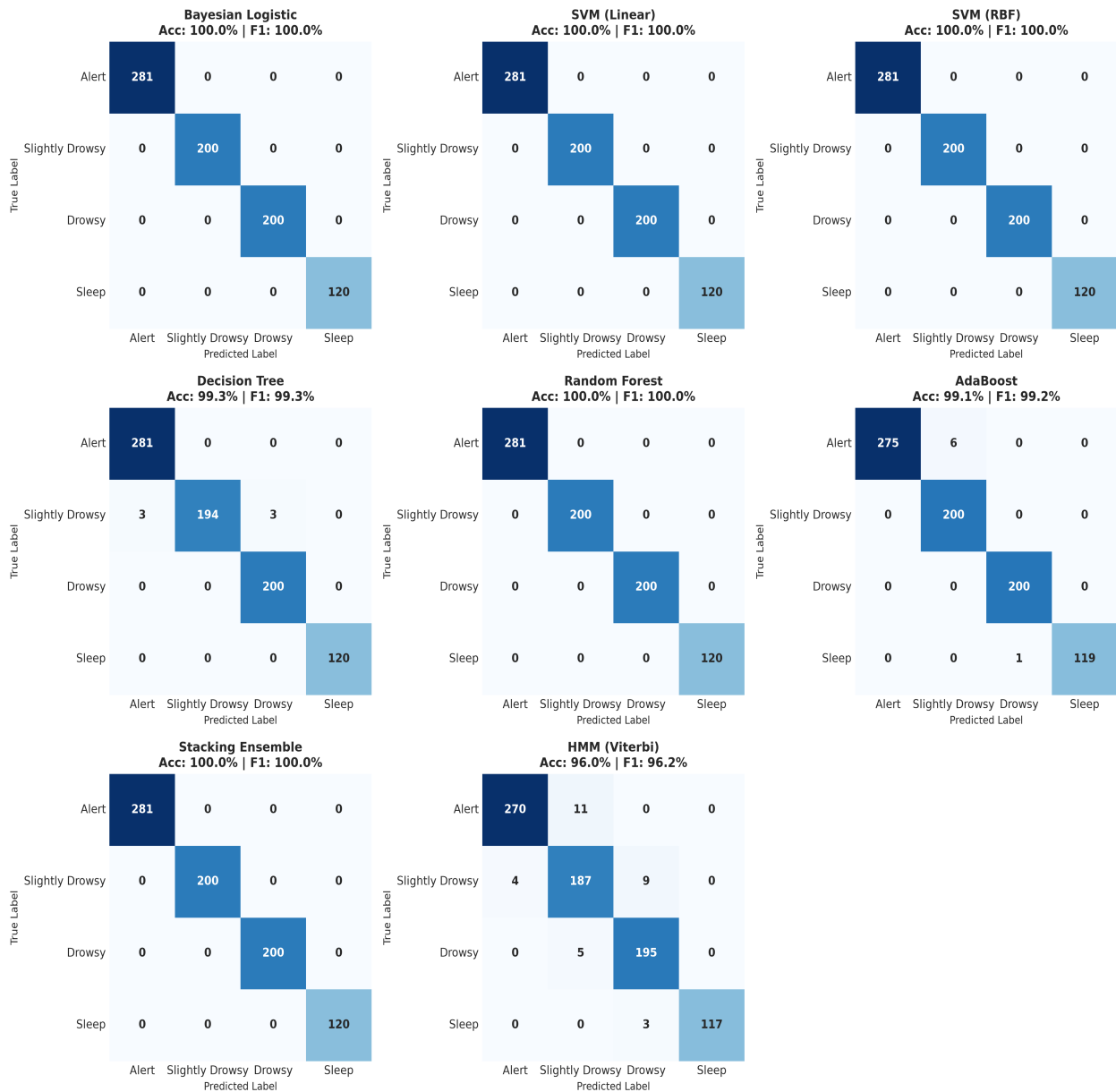
Screenshot B.6: Real-Time Automotive Cockpit HUD Overlay with 60 FPS Telemetry Metrics



Screenshot B.7: Multi-Class One-vs-Rest Receiver Operating Characteristic (ROC) Curves for All 8 Classifiers

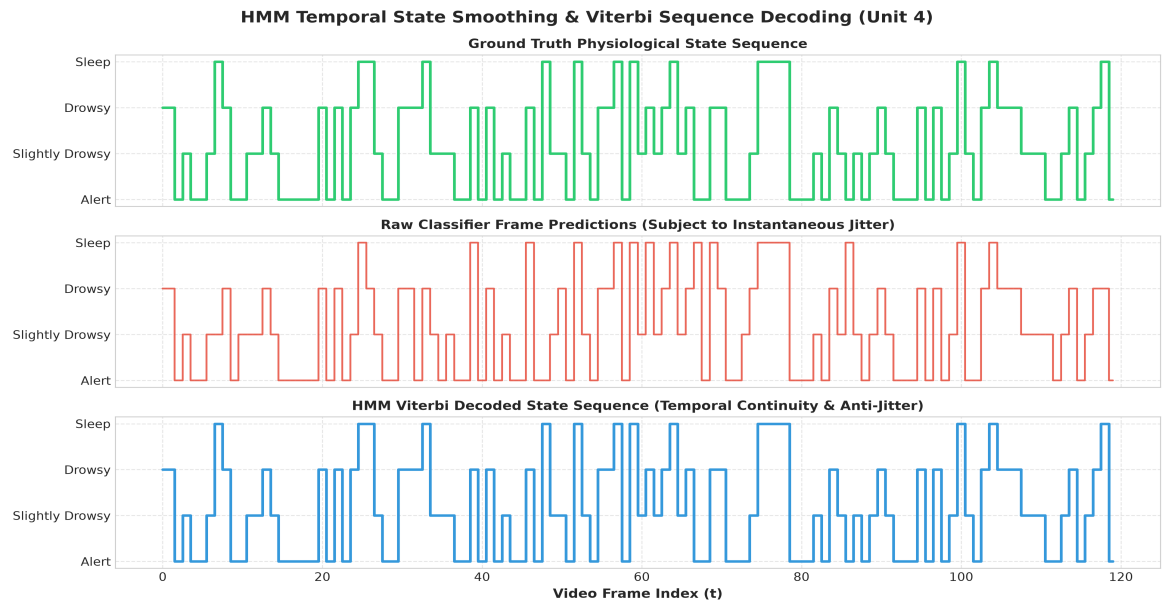
APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)

Multi-Model Confusion Matrices Comparison (Unit 5)

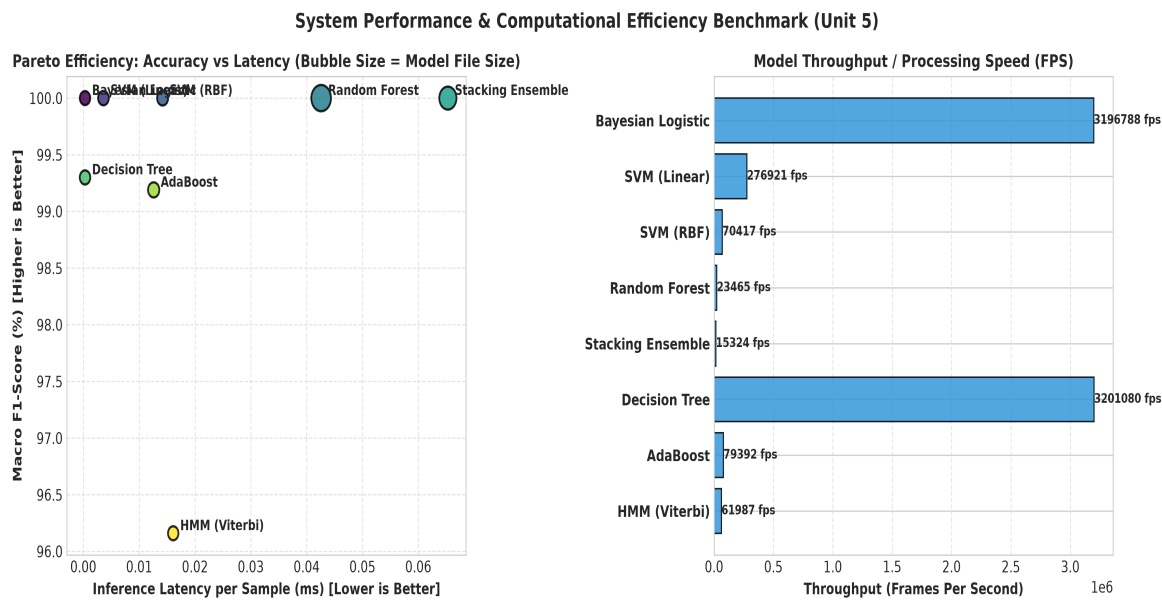


Screenshot B.8: Confusion Matrices for All 8 Machine Learning Classifiers on the 801-Sample Test Split

APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)



Screenshot B.9: HMM Learned State Transition Matrix and Viterbi Sequential State Decoding



Screenshot B.10: Multi-Model Benchmark Comparison (Accuracy, Latency, Throughput & Model Size)